4 **(a)** State **two** ways that evaporation differs from boiling.

1 .....

2 .....

[2]

(b) Fig. 4.1 shows part of a container used to store a mixture of liquid and gaseous oxygen.

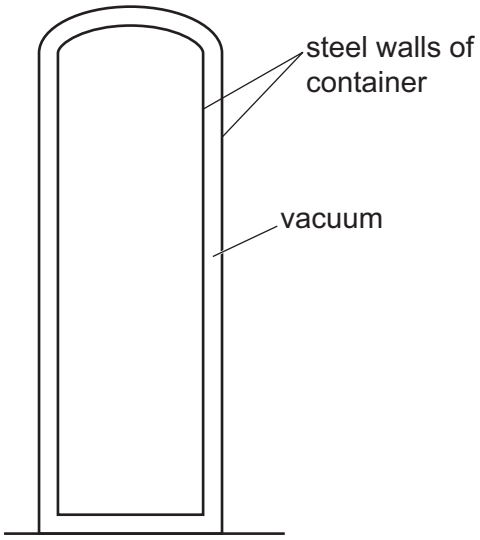

Fig. 4.1

The temperature of the liquid oxygen is –160 °C.

(i) Determine the temperature of the liquid oxygen in K.

temperature = ... K [1]

(ii) The container is made of steel and there is a vacuum between the inner and outer walls. The outer wall of the container is at room temperature.

State **two** methods of thermal energy transfer that a vacuum prevents.

1 .....

2 .....

[2]

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(c) Describe, in terms of particles, how a gas exerts a pressure on the walls of its container.

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..... [3]

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